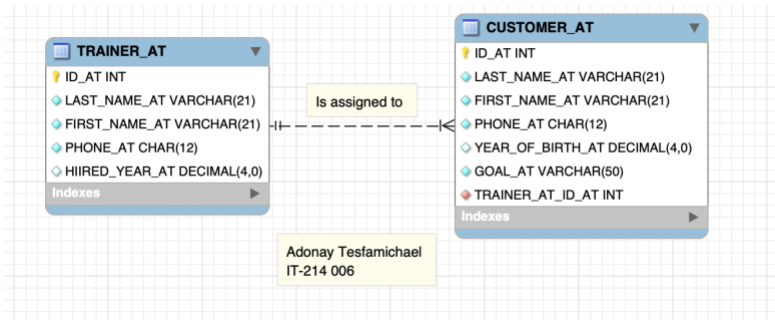


S2

Adonay Tesfamichael

IT214-006

Task1 - ERD Model



Name: TRAINER_AT						Name: CUSTOMER_AT					
Column	Datatype	PK	NN	UQ		Column	Datatype	PK	NN	UQ	
ID_AT	INT	☒	☒	☒		ID_AT	INT	☒	☒	☒	
LAST_NAME_AT	VARCHAR(21)	☐	☒	☐		LAST_NAME_AT	VARCHAR(21)	☐	☒	☐	
FIRST_NAME_AT	VARCHAR(21)	☐	☒	☐		FIRST_NAME_AT	VARCHAR(21)	☐	☒	☐	
PHONE_AT	CHAR(12)	☐	☒	☐		PHONE_AT	CHAR(12)	☐	☒	☐	
HIRED_YEAR_AT	DECIMAL(4,0)	☐	☐	☐		YEAR_OF_BIRTH_AT	DECIMAL(4,0)	☐	☐	☐	
						GOAL_AT	VARCHAR(50)	☐	☒	☐	
						TRAINER_AT_ID_AT	INT	☐	☒	☐	

Task2 - Create database by Adonay Tesfamichael

The screenshot shows the SQL Developer interface with the following SQL commands executed:

```
1 -- Task 2 Create database by Adonay Tesfamichael
2 * DROP DATABASE IF EXISTS FITNESS_COMPANY_AT;
3 * CREATE DATABASE FITNESS_COMPANY_AT;
4 * USE FITNESS_COMPANY_AT;
5 * SHOW DATABASES;
```

The **Result Grid** shows the output of the **SHOW DATABASES;** command, listing the following databases:

- fitness_company_at
- happy_learning_at
- happy_time
- hr_at
- information_schema
- mysql
- new_db
- performance_schema
- sys
- sys

The **Action Output** table shows the execution details:

	Time	Action	Response	Duration / Fetch Time
1	10:45:47	DROP DATABASE IF EXISTS FITNESS_COMPANY_AT	2 row(s) affected	0.073 sec
2	10:46:09	CREATE DATABASE FITNESS_COMPANY_AT	1 row(s) affected	0.0023 sec
3	10:46:13	USE FITNESS_COMPANY_AT	0 row(s) affected	0.00067 sec
4	10:46:15	SHOW DATABASES	11 row(s) returned	0.0023 sec / 0.00002...

Task3 - Create table by Adonay Tesfamichael

-- Task 3 - Create table by Adonay Tesfamichael

SHOW TABLES;

```
CREATE TABLE TRAINER_AT(  
ID_AT INT PRIMARY KEY NOT NULL,  
LAST_NAME_AT VARCHAR(21) NOT NULL,  
FIRST_NAME_AT VARCHAR(21) NOT NULL,  
PHONE_AT CHAR(12) NOT NULL,  
HIRED_YEAR_AT DECIMAL(4, 0) NULL  
);  
SHOW TABLES;
```

```
CREATE TABLE CUSTOMER_AT (  
ID_AT INT PRIMARY KEY NOT NULL,  
LAST_NAME_AT VARCHAR(21) NOT NULL,  
FIRST_NAME_AT VARCHAR(21) NOT NULL,  
PHONE_AT CHAR(12) NOT NULL,  
YEAR_OF_BIRTH_AT DECIMAL(4,0) NULL,  
GOAL_AT VARCHAR(50) NOT NULL,  
TRAINER_AT_ID_AT INT,  
FOREIGN KEY (TRAINER_AT_ID_AT) REFERENCES TRAINER_AT(ID_AT)  
);  
SHOW TABLES;
```

```
SELECT * FROM TRAINER_AT,CUSTOMER_AT;  
DESCRIBE TRAINER_AT;  
DESCRIBE CUSTOMER_AT;
```

TRAINER

CUSTOMER

21:32								1:39							
Result Grid								Result Grid							
Filter Rows: Search								Filter Rows: Search							
Export:								Export:							
Field	Type	Null	Key	Default	Extra			Field	Type	Null	Key	Default	Extra		
ID_AT	int	NO	PRI	NULL				ID_AT	int	NO	PRI	NULL			
LAST_NAME_AT	varchar(21)	NO		NULL				LAST_NAME_AT	varchar(21)	NO		NULL			
FIRST_NAME_AT	varchar(21)	NO		NULL				FIRST_NAME_AT	varchar(21)	NO		NULL			
PHONE_AT	char(12)	NO		NULL				PHONE_AT	char(12)	NO		NULL			
HIRED_YEAR_AT	decimal(4,0)	YES		NULL				YEAR_OF_BIRTH_AT	decimal(4,0)	YES		NULL			
								GOAL_AT	varchar(50)	NO		NULL			
								TRAINER_AT_ID_AT	int	YES	MUL	NULL			

Action Output			Response		
	Time	Action			
1	10:56:43	DROP DATABASE IF EXISTS FITNESS_COMPANY_AT		1 row(s) affected	
2	10:56:46	CREATE DATABASE FITNESS_COMPANY_AT		1 row(s) affected	
3	10:56:48	USE FITNESS_COMPANY_AT		0 row(s) affected	
4	10:56:50	SHOW DATABASES		11 row(s) returned	
5	10:56:54	SHOW TABLES		0 row(s) returned	
6	10:57:00	CREATE TABLE TRAINER_AT (ID_AT INT PRIMARY KEY NOT NULL, LAST_NAME_AT VARCHAR(21) NOT NULL, FIRST_NAME_AT VARCH...		0 row(s) affected	
7	10:57:08	SHOW TABLES		1 row(s) returned	
8	10:59:23	CREATE TABLE CUSTOMER_AT (ID_AT INT PRIMARY KEY NOT NULL, LAST_NAME_AT VARCHAR(21) NOT NULL, FIRST_NAME_AT VARCH...		0 row(s) affected	
9	10:59:27	SHOW TABLES		2 row(s) returned	
10	11:00:03	SELECT * FROM TRAINER_AT,CUSTOMER_AT LIMIT 0, 1000		0 row(s) returned	
11	11:00:45	DESCRIBE TRAINER_AT		5 row(s) returned	
12	11:00:48	DESCRIBE CUSTOMER_AT		7 row(s) returned	
13	11:02:23	DESCRIBE CUSTOMER_AT		7 row(s) returned	
14	11:02:27	DESCRIBE TRAINER_AT		5 row(s) returned	

Task4 - Add data by Adonay Tesfamichael

```
36 -- Task 4 - Add data by Adonay Tesfamichael
37 • INSERT INTO TRAINER_AT VALUES
38 (1, 'Jones', 'William', '175-221-9988', 2019),
39 (2, 'John', 'Tomus', '222-489-1234', 2014),
40 (3, 'Fillap', 'Ben', '175-223-9998', 2012),
41 (4, 'John', 'Mike', '175-672-8912', 2020),
42 (5, 'Cindy', 'Timms', '175-098-2678', NULL);
43
44 • INSERT INTO CUSTOMER_AT VALUES
45 (1, 'Garcia', 'Robert', '175-751-8822', 1988, 'Building endurance', 1),
46 (2, 'John', 'Sue', '827-738-9823', 2000, 'Loss weight', 3),
47 (3, 'Son', 'Woo', '175-898-0902', 1974, 'Run faster', 5 ),
48 (4, 'Rodus', 'Sue', '175-123-4566', 2004, 'Look her best, but not come a lot', 2),
49 (5, 'John', 'Bills', '175-272-7272', NULL, 'Get bigger', 3);
50
51 • SELECT * FROM TRAINER_AT;
52 • SELECT * FROM CUSTOMER_AT;
```

Trainer

ID_AT	LAST_NAME_AT	FIRST_NAME_AT	PHONE_AT	HIRED_YEAR_AT
1	Jones	William	175-221-9988	2019
2	John	Tomus	222-489-1234	2014
3	Fillap	Ben	175-223-9998	2012
4	John	Mike	175-672-8912	2020
5	Cindy	Timms	175-098-2678	NULL
NULL	NULL	NULL	NULL	NULL

Customer

ID_AT	LAST_NAME_AT	FIRST_NAME_AT	PHONE_AT	YEAR_OF_BIRTH_AT	GOAL_AT	TRAINER_AT_ID_AT
1	Garcia	Robert	175-751-8822	1988	Building endurance	1
2	John	Sue	827-738-9823	2000	Loss weight	3
3	Son	Woo	175-898-0902	1974	Run faster	5
4	Rodus	Sue	175-123-4566	2004	Look her best, but not come a lot	2
5	John	Bills	175-272-7272	NULL	Get bigger	3
NULL	NULL	NULL	NULL	NULL	NULL	NULL

Time	Action	Response
10:56:43	SHOW DATABASE IF EXISTS FITNESS_COMPANY_AT	1 row(s) affected
10:56:48	CREATE DATABASE FITNESS_COMPANY_AT	1 row(s) affected
10:56:48	USE FITNESS_COMPANY_AT	0 row(s) affected
10:56:50	SHOW DATABASES	11 row(s) returned
10:56:54	SHOW TABLES	0 row(s) returned
10:57:00	CREATE TABLE TRAINER_AT (ID_AT INT PRIMARY KEY NOT NULL, LAST_NAME_AT VARCHAR(20) NOT NULL, FIRST_NAME_AT VARCHAR(20) NOT NULL, PHONE_AT VARCHAR(20) NOT NULL, HIRED_YEAR_AT INT NOT NULL)	0 row(s) affected
10:57:08	SHOW TABLES	1 row(s) returned
10:59:23	CREATE TABLE CUSTOMER_AT (ID_AT INT PRIMARY KEY NOT NULL, LAST_NAME_AT VARCHAR(20) NOT NULL, FIRST_NAME_AT VARCHAR(20) NOT NULL, YEAR_OF_BIRTH_AT INT NOT NULL, GOAL_AT VARCHAR(50) NOT NULL, TRAINER_AT_ID_AT INT NOT NULL)	1 row(s) affected
10:59:27	SHOW TABLES	2 row(s) returned
11:00:03	SELECT * FROM TRAINER_AT LIMIT 0, 1000	0 row(s) returned
11:00:41	DESCRIBE TRAINER_AT	6 row(s) returned
11:00:48	DESCRIBE CUSTOMER_AT	7 row(s) returned
11:02:23	DESCRIBE CUSTOMER_AT	7 row(s) returned
11:02:27	DESCRIBE TRAINER_AT	6 row(s) returned
11:15:15	INSERT INTO TRAINER_AT VALUES (1, 'Jones', 'William', '175-221-9988', 2019), (2, 'John', 'Tomus', '222-489-1234', 2014), (3, 'Fillap', 'Ben', '175-223-9998', 2012), (4, 'John', 'Mike', '175-672-8912', 2020), (5, 'Cindy', 'Timms', '175-098-2678', NULL)	5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0
11:15:20	SELECT * FROM TRAINER_AT LIMIT 0, 1000	5 row(s) returned
11:16:12	INSERT INTO CUSTOMER_AT VALUES (1, 'Garcia', 'Robert', '175-751-8822', 1988, 'Building endurance', 1), (2, 'John', 'Sue', '827-738-9823', 2000, 'Loss weight', 3), (3, 'Son', 'Woo', '175-898-0902', 1974, 'Run faster', 5), (4, 'Rodus', 'Sue', '175-123-4566', 2004, 'Look her best, but not come a lot', 2), (5, 'John', 'Bills', '175-272-7272', NULL, 'Get bigger', 3)	5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0
11:16:16	SELECT * FROM CUSTOMER_AT LIMIT 0, 1000	5 row(s) returned
11:16:56	SELECT * FROM TRAINER_AT LIMIT 0, 1000	5 row(s) returned
11:17:40	SELECT * FROM CUSTOMER_AT LIMIT 0, 1000	5 row(s) returned

Task5

- For each assignment, list the first and last names for both the trainer and the customer.

Use suggestive names for the columns in the result.

```

57 -- For each assignment, list the first and last names for both the trainer and the customer. Use suggestive names for the columns in the result.
58 • SELECT T.FIRST_NAME_AT AS 'TRAINERS FIRST NAME',
59       T.LAST_NAME_AT AS 'TRAINERS LAST NAME',
60       C.FIRST_NAME_AT AS 'CUSTOMER FIRST NAME',
61       C.LAST_NAME_AT AS 'CUSTOMER LAST NAME'
62 FROM CUSTOMER_AT C, TRAINER_AT T
63 WHERE C.TRAINER_AT_ID_AT = T.ID_AT;

```

5% 36:63

Result Grid Filter Rows: Search Export:

TRAINERS FIRST NAME	TRAINERS LAST NAME	CUSTOMER FIRST NAME	CUSTOMER LAST NAME
William	Jones	Robert	Garcia
Ben	Fillap	Sue	John
Timms	Cindy	Woo	Son
Tomus	John	Sue	Rodus
Ben	Fillap	Bills	John

- Pick the ID for the third trainer you defined in the database. List the full names (FIRST SPACE LAST) for all the customers of the selected trainer (in one column named "Customer").

```

65 -- Pick the ID for the third trainer you defined in the database. List the full names (FIRST SPACE LAST) for all the customers of the selected trainer (in one column named "Customer").
66 • SELECT CONCAT(FIRST_NAME_AT, ' ', LAST_NAME_AT) AS 'FULL NAME'
67 FROM CUSTOMER_AT
68 WHERE TRAINER_AT_ID_AT = 3;

```

75% 28:68

Result Grid Filter Rows: Search Export:

FULL NAME
Sue John
Bills John

- List the full names (FIRST SPACE LAST) for the trainer and customers (in two columns named "Trainer" and "Customer"), for which the customer first name is "Sue"

```

70 -- List the full names (FIRST SPACE LAST) for the trainer and customers (in two columns named "Trainer" and "Customer"), for which the customer first name is "Sue"
71 • SELECT CONCAT(T.FIRST_NAME_AT, ' ', T.LAST_NAME_AT) AS 'Trainer',
72       CONCAT(C.FIRST_NAME_AT, ' ', C.LAST_NAME_AT) AS 'Customer'
73 FROM CUSTOMER_AT C
74 JOIN TRAINER_AT T ON C.TRAINER_AT_ID_AT = T.ID_AT
75 WHERE C.FIRST_NAME_AT = 'SUE';

```

5% 31:75

Result Grid Filter Rows: Search Export:

Trainer	Customer
Ben Fillap	Sue John
Tomus John	Sue Rodus

- List the names for the trainers in the format (LAST COMMA SPACE FIRST) hired after 2015 and the full names of their customers. Show the result in two columns named "Trainer" and "Customer".

```
77 -- List the names for the trainers in the format (LAST COMMA SPACE FIRST) hired after 2015 and the full names of their customers. Show the result in two columns named "Trainer" and "Customer"
78 SELECT CONCAT(T.LAST_NAME_AT, ', ', T.FIRST_NAME_AT) AS 'Trainer',
79        CONCAT(C.LAST_NAME_AT, ', ', C.FIRST_NAME_AT) AS 'Customer'
80 FROM CUSTOMER_AT C
81 JOIN TRAINER_AT T ON C.TRAINER_AT_ID_AT= T.ID_AT
82 WHERE T.HIRED_YEAR_AT > 2015
83
```

Result Grid

Trainer	Customer
Jones, William	Garcia Robert

Action Output

	Time	Action	Response	Duration / Fetch Time
✓ 1	16:44:27	DROP DATABASE IF EXISTS FITNESS_COMPANY_AT	2 row(s) affected	0.064 sec
✓ 2	16:44:27	CREATE DATABASE FITNESS_COMPANY_AT	1 row(s) affected	0.0010 sec
✓ 3	16:44:27	USE FITNESS_COMPANY_AT	0 row(s) affected	0.00027 sec
✓ 4	16:44:27	SHOW DATABASES	11 row(s) returned	0.0092 sec / 0.00000...
✓ 5	16:44:27	SHOW TABLES	0 row(s) returned	0.0031 sec / 0.00000...
✓ 6	16:44:27	CREATE TABLE TRAINER_AT (ID_AT INT PRIMARY KEY NOT NULL, LAST_NAME_AT VARCHAR(21) NOT NULL, FIRST_NAME_AT VARCH...	0 row(s) affected	0.012 sec
✓ 7	16:44:27	SHOW TABLES	1 row(s) returned	0.0012 sec / 0.00001...
✓ 8	16:44:27	CREATE TABLE CUSTOMER_AT (ID_AT INT PRIMARY KEY NOT NULL, LAST_NAME_AT VARCHAR(21) NOT NULL, FIRST_NAME_AT VAR...	0 row(s) affected	0.0089 sec
✓ 9	16:44:27	SHOW TABLES	2 row(s) returned	0.00098 sec / 0.000...
✓ 10	16:44:28	SELECT * FROM TRAINER_AT,CUSTOMER_AT LIMIT 0, 1000	0 row(s) returned	0.0029 sec / 0.00001...
✓ 11	16:44:28	DESCRIBE TRAINER_AT	5 row(s) returned	0.0041 sec / 0.00001...
✓ 12	16:44:28	DESCRIBE CUSTOMER_AT	7 row(s) returned	0.0013 sec / 0.00001...
✓ 13	16:44:28	INSERT INTO TRAINER_AT VALUES (1, 'Jones', 'William', '175-221-9988', 2019), (2, 'John', 'Tomus', '222-489-1234', 2014), (3, 'Fillap', '...	5 row(s) affected Records: 5 Duplicates: 0 Warnings...	0.0012 sec
✓ 14	16:44:28	INSERT INTO CUSTOMER_AT VALUES (1, 'Garcia', 'Robert', '175-751-8822', 1988, 'Building endurance', 1), (2, 'John', 'Sue', '827-738-98...	5 row(s) affected Records: 5 Duplicates: 0 Warnings...	0.00081 sec
✓ 15	16:44:28	SELECT * FROM TRAINER_AT LIMIT 0, 1000	5 row(s) returned	0.00032 sec / 0.0000...
✓ 16	16:44:28	SELECT * FROM CUSTOMER_AT LIMIT 0, 1000	5 row(s) returned	0.00041 sec / 0.0000...
✓ 17	16:44:28	SELECT T.FIRST_NAME_AT AS 'TRAINERS FIRST NAME', T.LAST_NAME_AT AS 'TRAINERS LAST NAME', C.FIRST_NAME_AT AS '...	5 row(s) returned	0.00048 sec / 0.000...
✓ 18	16:44:28	SELECT CONCAT(FIRST_NAME_AT, ', ', LAST_NAME_AT) AS 'FULL NAME' FROM CUSTOMER_AT WHERE TRAINER_AT_ID_AT = 3 LIMIT...	2 row(s) returned	0.00049 sec / 0.000...
✓ 19	16:44:28	SELECT CONCAT(T.FIRST_NAME_AT, ', ', T.LAST_NAME_AT) AS 'Trainer', CONCAT(C.FIRST_NAME_AT, ', ', C.LAST_NAME_AT) AS 'Cu...	2 row(s) returned	0.00048 sec / 0.000...
✓ 20	16:44:28	SELECT CONCAT(T.LAST_NAME_AT, ', ', T.FIRST_NAME_AT) AS 'Trainer', CONCAT(C.LAST_NAME_AT, ', ', C.FIRST_NAME_AT) AS 'Cus...	1 row(s) returned	0.00073 sec / 0.0000...